Write an 8086 Assembly Language Program to Perform Multiplication of Two Numbers Using the MUL Instruction.

```
org 100h

mov ax,00h

mov bx,04h

mov cx,02h

L1:add ax,bx

dec cx

jnz L1

ret
```

Write an 8086 Assembly Language Program to Calculate the Factorial of a Number Using the MUL Instruction.

```
org 100h

mov bx,04h

mov ax,01h

L1:mul bx

dec bx

jnz L1

ret
```

Write an 8086 Assembly Language Program to Draw a Square in Graphics Mode Using BIOS Interrupt INT 10H.

```
org 100h


mov ax,0013h

int 10h


mov bx,60h
```

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```
mov cx,60h
mov dx,60h

mov al,02h
l1:mov ah,0ch
inc cx
dec bx
int 10h
jnz l1

mov bx,60h
mov al,22h
l2:mov ah,0ch
inc dx
dec bx
int 10h
jnz l2

mov bx,60h
mov al,02h
l3:mov ah,0ch
dec cx
dec bx
int 10h
jnz l3

mov bx,60h
mov al,22h
```

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```
l4:mov ah,0ch
dec dx
dec bx
int 10h
jnz l4

ret
```

Write an 8086 Assembly Language Program to Display Uppercase and Lowercase Alphabets (A–Z and a–z) Using DOS Interrupt 21H.

```
Mov ah,02h                  Write a single character to standard output
Mov cx,26
Mov dl,'A'
l1:int 21h
Inc dl
Dec cx
Jnz l1

Mov dl,0ah                  Line feed (moves to next output line)
Int 21h

Mov dl,0dh                  Carriage return (moves to leftmost output column)
Int 21h

Mov ah,02h
Mov cx,26
```

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```
Mov dl,’a’
L2:int 21h
Inc dl
Dec cx
Jnz L2
```

Write an 8086 Assembly Language Program to Convert an Uppercase String into a Lowercase String.org 100h

```
.data
m1 db 10,13,'Enter the string:$'
m2 db 10,13,'The lowercase string:$'

buff db 80

.code
mov ah,09h
lea dx,m1
int 21h

mov ah,0ah
lea dx,buff
int 21h

mov ah,09h
```

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```
lea dx,m2
int 21h

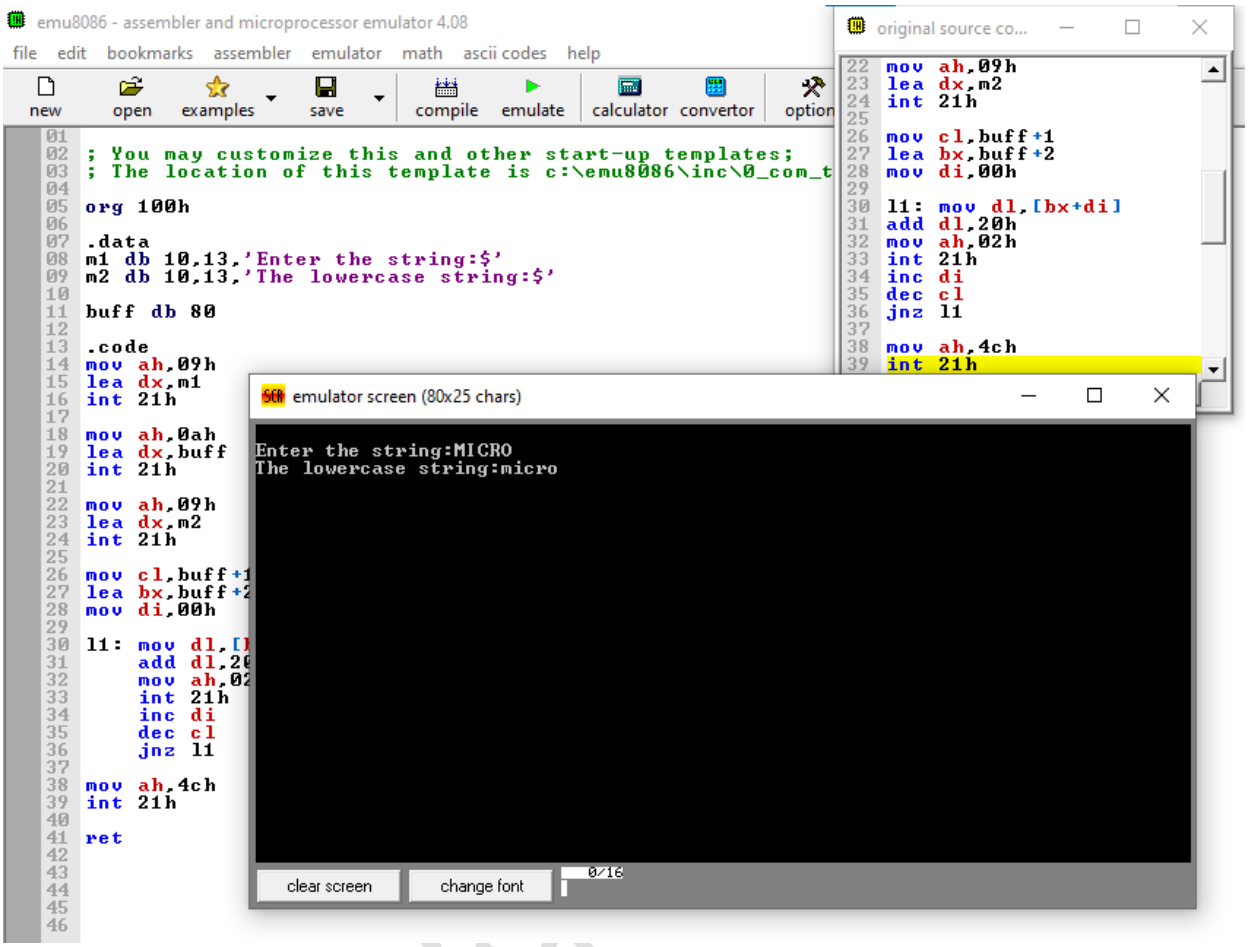
mov cl,buff+1
lea bx,buff+2
mov di,00h

l1: mov dl,[bx+di]
    add dl,20h
    mov ah,02h
    int 21h
    inc di
    dec cl
    jnz l1

mov ah,4ch
int 21h

ret
```

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Write an 8086 Assembly Language Program to Convert a Lowercase String into an Uppercase String.

org 100h

.data

m1 db 10,13,'Enter the string:$'

m2 db 10,13,'The uppercase string:$'

buff db 80

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```
.code

mov ah,09h
lea dx,m1
int 21h

mov ah,0ah
lea dx,buff
int 21h

mov ah,09h
lea dx,m2
int 21h

mov cl,buff+1
lea bx,buff+2
mov di,00h

l1: mov dl,[bx+di]
    sub dl,20h
    mov ah,02h
    int 21h
    inc di
    dec cl
```

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```
    jnz l1

mov ah,4ch
int 21h

ret
```

Write an 8086 Assembly Language Program to Swap Two Numbers Using a Temporary Register.

```
org 100h

mov ax,1234h
mov bx,5678h

mov cx,ax
mov ax,bx
mov bx,cx

ret
```

Write an 8086 Assembly Language Program to Transfer a Block of Data from One Memory Location to Another Using the MOVSB Instruction.

```
org 100h

.data
```

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```
src db 10h,20h,30h,40h,50h
dest db 5 dup(0)

.code

lea si,src
lea di,dest
mov cx,05h
cld

l1: movsb
    loop l1

ret
```

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